

Cybersecurity, Linux, Networking & Cryptography Notes

These notes provide a beginner-friendly yet detailed overview of key cybersecurity concepts, Linux fundamentals, networking basics, and cryptography principles.

1. CIA Triad

Principle	Meaning	Example
Confidentiality	Protecting data from unauthorized access	Passwords, encryption
Integrity	Ensuring data remains accurate and unchanged	Hashing, digital signatures
Availability	Ensuring systems/services remain accessible	Backups, redundancy

2. Common Threat Types

Phishing: Fraudulent emails/messages designed to steal credentials.

Malware: Malicious software such as viruses, worms, and trojans.

DDoS: Distributed Denial of Service attacks overwhelm servers with traffic.

SQL Injection: Attackers inject SQL queries into vulnerable applications.

Brute Force: Repeated attempts to guess passwords or keys.

Ransomware: Malware that encrypts files and demands payment.

3. Attack Vectors

- Social Engineering: Manipulating users into revealing confidential information.
- Wireless Attacks: Exploiting weak Wi-Fi security and rogue access points.
- Insider Threats: Threats caused by employees or trusted individuals.

4. Linux Fundamentals

Command	Purpose	Example
pwd	Show current directory	pwd
ls	List files/directories	ls -la
cd	Change directory	cd /home/user
chmod	Change permissions	chmod 755 file.txt

chown	Change ownership	chown user:user file.txt
apt	Package manager	sudo apt install nginx
dpkg	Manage .deb packages	dpkg -i package.deb
ifconfig	Display network interfaces	ifconfig
ping	Check connectivity	ping google.com
netstat	View network connections	netstat -tulnp
traceroute	Trace packet path	traceroute google.com

5. Networking Basics

OSI Model Layers

Layer	Function
Application	User-facing network services
Presentation	Data formatting and encryption
Session	Session management
Transport	Reliable data transfer (TCP/UDP)
Network	Routing and IP addressing
Data Link	MAC addressing and switching
Physical	Transmission of raw bits

TCP/IP Protocol Suite:

- Application Layer – HTTP, FTP, DNS
- Transport Layer – TCP and UDP
- Internet Layer – IP, ICMP
- Network Access Layer – Ethernet, Wi-Fi

DNS: Converts domain names into IP addresses.

HTTP: HyperText Transfer Protocol used for web communication.

HTTPS: Secure HTTP using SSL/TLS encryption.

IP Addressing: IPv4 uses 32-bit addresses while IPv6 uses 128-bit addresses.

Subnetting: Divides networks into smaller sub-networks.

NAT (Network Address Translation): Allows private IPs to communicate with public networks.

6. Cryptography Basics

Symmetric Encryption: Same key is used for encryption and decryption. Example: AES.

Asymmetric Encryption: Uses a public key and private key pair. Example: RSA.

Hashing: One-way conversion of data into a fixed-size value.

- MD5 – Fast but insecure for modern use.
- SHA256 – More secure and commonly used today.

Digital Certificates: Verify identity online and are issued by Certificate Authorities (CAs).

SSL/TLS: Protocols that encrypt communication between client and server.

7. OpenSSL Hands-on Commands

1 Generate RSA Private Key:

```
openssl genrsa -out private.key 2048
```

2 Extract Public Key:

```
openssl rsa -in private.key -pubout -out public.key
```

3 Encrypt a File:

```
openssl rsautl -encrypt -inkey public.key -pubin -in message.txt -out encrypted.bin
```

4 Decrypt a File:

```
openssl rsautl -decrypt -inkey private.key -in encrypted.bin -out decrypted.txt
```

End of Notes — Practice Linux commands, networking concepts, and cryptography hands-on to strengthen your cybersecurity fundamentals.